

TCP/IP Model and Protocols

Subject: Computer Networks Topic: Transport Layer

Overview

Four-layer model, TCP vs UDP, three-way handshake, congestion control

Four-layer model

- Application protocols such as HTTP, DNS, and SMTP use transport services.
- Transport layer TCP provides reliable ordered byte streams; UDP provides datagrams without built-in delivery guarantees.
- Internet layer IP addresses and forwards packets across networks; link layer handles delivery on a local medium.

TCP connection and reliability

- The three-way handshake is SYN, SYN-ACK, ACK; it synchronizes sequence numbers and establishes connection state.
- TCP numbers bytes, acknowledges received data, retransmits lost segments, and uses a receive window for flow control.
- Congestion control adjusts sending behavior based on network congestion; flow control protects the receiver.

TCP versus UDP

- Use TCP when reliable ordered delivery is useful, such as web pages and file transfer.
- UDP suits applications that implement their own timing or loss handling, such as real-time voice or some DNS queries.
- Neither protocol is universally faster; measure latency, overhead, and application requirements.

Troubleshooting

- DNS resolves names to records; a DNS failure can prevent a connection before TCP starts.
- A successful TCP handshake does not prove that an application request or TLS negotiation succeeded.
- Use packet captures to distinguish name resolution, routing, handshake, retransmission, and application-layer failures.